

Assignment - Linear & Logistic Regression

Course book: Machine Learning Algorithms (Linear Regression & Logistic Regression chapters). Answer all five questions. Your answers are graded against the book.

Q1. What does the coefficient of determination (R^2) measure, and what do values close to 1 versus close to 0 (or negative) tell you about the model?

Q2. Compare Ridge, Lasso, and ElasticNet regularization. For each, state which norm of the weights it penalizes and its characteristic effect on the weights.

Q3. What problem does RANSAC address in linear regression, and how does it work at a high level?

Q4. Why is logistic regression considered a classification method despite its name, and what is the role of the sigmoid (logistic) function?

Q5. In scikit-learn's LogisticRegression, what does the parameter C represent, and how do larger versus smaller values change the model?